Stanley Nelavala

Data Engineer: Billing, Payments and Go-to-Market Data Systems

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Huntington, IN | Open to relocate anywhere

Summary

Software engineer with an M.S. in Computer Science (4.0 GPA) and a year at Mass Technologies, a 10-person startup, where I build the RxApps360 hospital platform: FastAPI and PostgreSQL services, React and TypeScript web apps, a React Native mobile app, and the signed API every module calls. I work with data end to end: keyset paging and SQL tuning on a 328,000-row catalog, automated server and database schedules, and a public FHIR pipeline (Airflow, dbt, DuckDB) with data contracts, a quarantine gate, k-anonymity and 108 tests. I use Claude Code, Codex and opencode every day. AWS Certified Solutions Architect, Associate.

Experience

Software Development Engineer Nov 2025 – Present
Mass Technologies, team of 10 (RxApps360 hospital platform, pre-launch; numbers from dev and test) *Huntington, IN*

- Replaced offset paging with keyset paging on the catalog device list: row 150,000 of a 328,000-device dev database fell from 285 ms to 0.20 ms, with no skipped or repeated rows (tested with inserts, deletes and ties).
- Automated weekday start and stop of dev and acceptance servers and databases (ECS, RDS): about 53 hours a week instead of 168.
- Designed, built and integrated into every platform module the signed service-to-service API (M2M): each call proves which service and tenant sent it. After a security audit, hardened it with a Redis rate limiter and replay protection.
- Standardized malware scanning of file uploads (GuardDuty) across 9 app modules.
- Built RxDesign, the hospital design service (FastAPI, React, TypeScript), with a deterministic, rule-based layout engine and an LLM setting; 220+ offline API tests run with fake models, so tests never call an LLM.
- Fixed a regular expression denial-of-service bug in a shared validation package: an 80-character hostile input took about 138 s; the new linear pattern matched the old results on 60,000 random inputs.
- Use Claude Code, Codex and opencode every day to design, write, debug and review code; tests and written acceptance checks gate every change.

Adjunct Faculty (part-time) Jan 2026 – Aug 2026
The College of St. Scholastica *Duluth, MN*

- Taught Programming II (Java) and Machine Learning for Data Science.

Graduate Research Assistant and Teaching Assistant Aug 2023 – Aug 2025
University of Minnesota Duluth *Duluth, MN*

- Built a pipeline (Python, Node.js, MySQL, FFmpeg, PaddleOCR, BERTopic) that analyzed 1M+ TikTok videos from 300k+ authors for thesis research on misinformation.

AI projects (public code, tests)

FHIR Analytics Pipeline | *Python, Airflow, dbt, DuckDB, HL7 FHIR R4* GitHub

- Health-data pipeline with dbt model contracts, a quarantine gate, k-anonymity (k=5), HIPAA-style control mapping and 108 tests.

Payments Ledger Service | *Java, Spring Boot, PostgreSQL, React, TypeScript* GitHub

- Double-entry ledger with idempotency keys and an outbox; property tests and a load check that no money is created or lost.

Agent Integration Platform | *Python, FastAPI, LangGraph, MCP, PostgreSQL (pgvector), Redis* GitHub

- Multi-step agent with typed state, one constrained MCP write tool, human approval that resumes after restart, and step and token budgets; hybrid retrieval with access labels.
- 62 deterministic replay evals, prompt-injection tests, output guardrails, PII redaction, an append-only audit log; failover between two LLM adapters with a circuit breaker.

Skills

Role focus: billing and payments data, data contracts and reconciliation, internal systems, SQL and Python pipelines
AI-assisted coding: Claude Code and subagents, Codex, opencode, a DeepSeek harness; planner, manager and worker runs with acceptance checks
Data and SQL: SQL, PostgreSQL, dbt, Airflow, DuckDB, data contracts, data-quality gates, keyset paging, HL7 FHIR R4
Languages: Python, SQL, Java, TypeScript, Go, Rust
Cloud and quality: AWS (ECS, RDS, S3), Docker, GitHub Actions, pytest, dbt tests, Prometheus

Education and Certification

M.S. Computer Science, University of Minnesota Duluth, 2023 – 2025, GPA 4.0, Graduate Summer Research Fellowship 2024 and 2025
B.Tech Computer Science and Engineering, IIIT D&M Kancheepuram, 2019 – 2023
AWS Certified Solutions Architect, Associate, Feb 2025 – Feb 2028 (verifiable on Credly)